

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test - Kaprekar Contest

(NMTC- at **SUB-JUNIOR LEVEL** VII to VIII Standards)

Saturday, 29th August 2009.

Note:

- 1) Fill in the response sheet with your Name, Class, the institution through which you appear in the specified places.
- 2) Diagrams are only visual aids; they are not drawn to scale.
- 3) You are free to do rough work on separate sheets.
- 4) Duration of the test: 2 p.m. to 4. p.m.- 2 hours.

PART – A

Note:

1. Only one of the choices A,B,C,D is correct for each question. Shade that alphabet of your choice in the response sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).
2. For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{4}$ mark.

1. The number of three digit numbers whose digits are even is

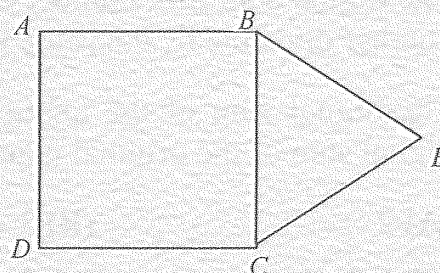
(A) 64

(B) 75

(C) 100

(D) 125

2. ABCD is a square and BCE is an equilateral triangle constructed externally. The measure of $\angle AED$ is



(A) 30°

(B) 20°

(C) 35°

(D) 15°

3. The number $111 \dots 1111$ is a 2009 digit number. It is multiplied by 2009. The third digit from the left of the product is

(A) 1

(B) 2

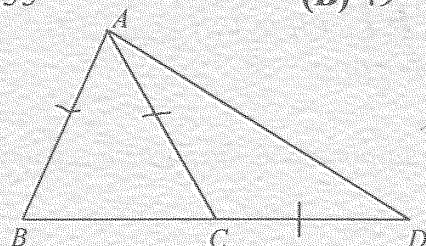
(C) 3

(D) 9

4. The average of ten different positive integers is 10. The smallest is 5. The biggest of these numbers can be

(A) 55 (B) 49 (C) 25 (D) 19

5.



In the figure $AB = AC = CD$ and

$\angle BAC = 32^\circ$, then $\angle BAD$ is

(A) 37° (B) 64° (C) 69° (D) 74°

6. The remainder when the number $(2 \times 3 \times 4 \times 2007 \times 2008 \times 2009) - 2008$ is divided by 2009 is

(A) 0 (B) 2008 (C) 1 (D) 2007

7. In a certain month in the new academic year, Sundays are on even dates in that month. Which day occurs on three even dates of the following month

(A) Thursday or Friday (B) Wednesday or Thursday
(C) Tuesday or Wednesday (D) Monday or Sunday

8. The digital sum of a number is the sum of the digits of the number. A three digit number divisible by 9 has the following property

The digital sum of the quotient of this number divided by 9 is 9 less than the digital sum of the original number. The number of such three digits numbers is

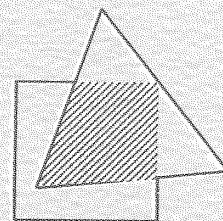
(A) 2 (B) 3 (C) 4 (D) 5

9. In the adjoining addition each letter represents a digit (identical letters represent the same digit, different letters represent different digits) The value of A is

$$\begin{array}{r} A A \\ + A A \\ \hline C A B \end{array}$$

(A) 6 (B) 7 (C) 8 (D) 9

10. A square and a triangle overlap as shown here. The shaded area is two thirds the area of the square and 40% of the area of the triangle. The ratio of the area of the square to the area of the triangle is



(A) 5:3 (B) 3:5 (C) 4:15 (D) 7:15

PART – B

Note:

- Write the correct answer in the space provided in the response sheet.
- For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{2}$ mark.

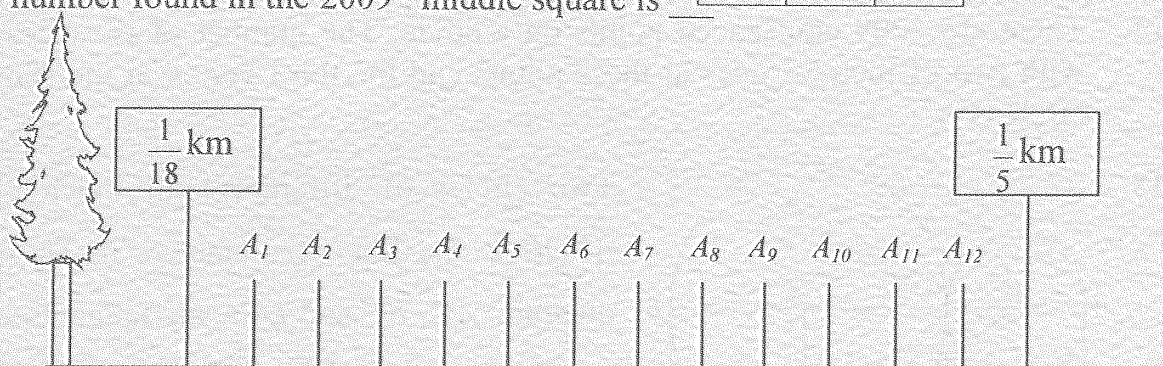
11. The natural numbers are filled as per the table shown where * represents a square not filled. We find in the middle squares the numbers

1	2	*
3	*	4
*	5	6
7	8	*
9	*	10
*	11	12
13	14	*
:	:	:
:	:	:

2, *, 5, 8, *,

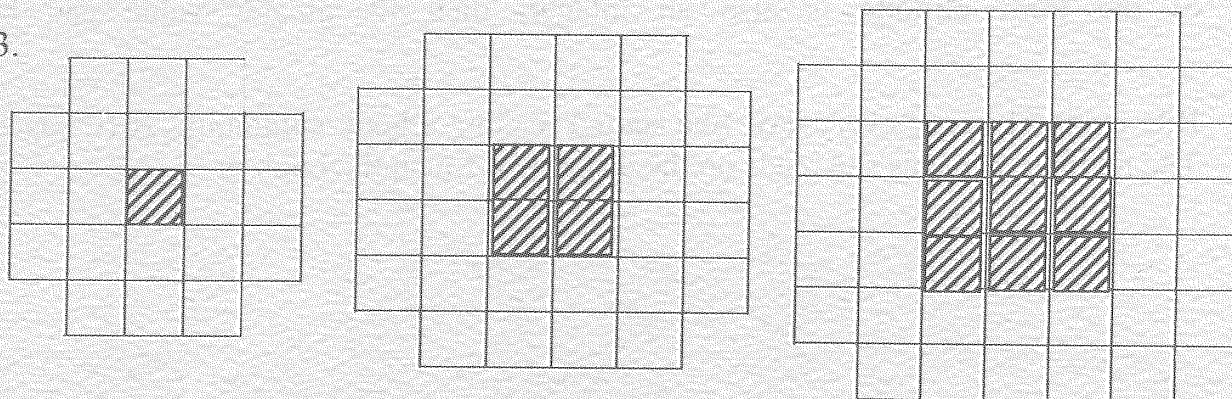
The number found in the 2009th middle square is _____

- 12.



In a straight road distances in kilometers are marked from a tree. Two of the sign boards show $\frac{1}{18}$ km, and $\frac{1}{5}$ km. Which position of the sign that shows $\frac{1}{6}$ km from the tree _____

- 13.

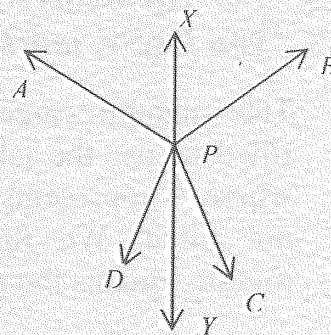


Observe the above diagram formed by unit squares. Not including the middle square hole, the number of unit squares needed to build the tenth pattern is _____

14. Three digit numbers are written with different digits. The sum of the biggest and the smallest of such numbers is _____

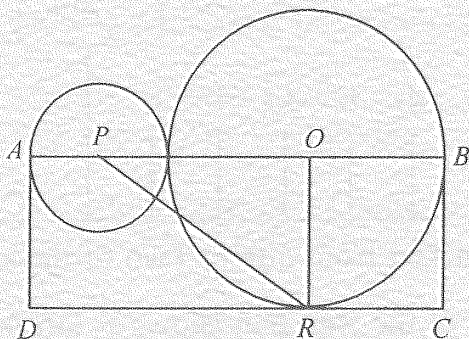
15. A certain number ' n ' is divisible by 21, 28 and 49. The smallest possible value of n is _____

16. In the diagram $\angle APB$ and $\angle DPC$ have a common vertex P . XPY is the common bisector of the angles $\angle APB$ and $\angle DPC$. If $\angle APB = 150^\circ$, $\angle CPD = 40^\circ$ then $\angle BPD$ is _____



17. There are 2009 equispaced points on a circle. The number of isosceles triangles with two of their vertices at these points and the third vertex at the centre of the circle is _____

18.



Two circles of different radii touch externally. $APQB$ is the line of diameters as shown in the diagram. AD , BC and DRC are tangents so that the area of the rectangle $ABCD$ is 16cm^2 . The area of the triangle PQR is _____ cm^2 .

19. Two of the altitudes of the scalene triangle ABC have length 4 and 12. If the length of the third altitude is also an integer, then its greater value can be _____

20. Rite reads 41 pages every day. She started reading a book containing 2010 pages. The number of pages she reads on the last day to finish the book is _____